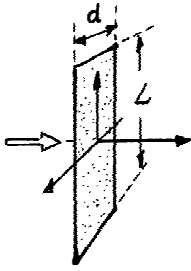
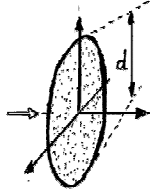
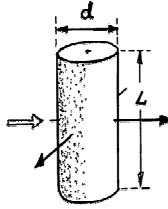
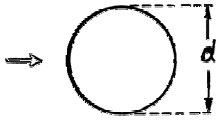
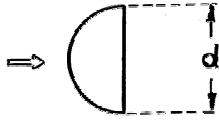


F: Table of drag coefficients

	C_d	Object and Range for R_{Rey}
	$\frac{L}{d}$: 1 1.18 5 1.2 10 1.3 20 1.5 30 1.6 ∞ 1.95	Flat Plate $10^3 < R_{Rey}$
	1.17	Circular Disk $10^3 < R_{Rey}$
	$\frac{L}{d}$: 1 0.63 5 0.80 10 0.83 20 0.93 30 1.0 ∞ 1.2	Circular Cylinder $10^3 < R_{Rey} < 10^5$
	0.47	Sphere $10^3 < R_{Rey} < 3 \cdot 10^5$
	0.42	Solid Hemisphere $10^4 < R_{Rey} < 10^6$
	1.17	$10^4 < R_{Rey} < 10^6$
	0.34	Hollow Hemisphere $10^4 < R_{Rey} < 10^6$
	1.42	$10^4 < R_{Rey} < 10^6$

The dimensionless drag coefficient C_D in the expression

$$F = \frac{1}{2} \rho A v^2 C_d \quad (F1)$$

for the force F against a body with frontal area A which moves with a velocity v ($< v_{\text{sound}}$) through a medium with mass density ρ . The values are from [11 p. 85]. As explained more detailed in sec. 8, C_d is a function of the dimensionless *Reynolds number*

$$R_{Rey} = \frac{\rho v}{\eta \lambda} \quad (F2)$$

where η is the viscosity of the medium and λ is the characteristic length chosen to characterize the size of the body in question. In all the examples considered in the table, the choice made is

$$\lambda = d \quad (F3)$$

Writing Reynolds number as

$$R_{Rey} = \frac{v}{v_{Rey}}, \quad (F4)$$

it can be considered as the velocity v measured in units of "Reynolds velocity"

$$v_{Rey} = \frac{\eta \lambda}{\rho}, \quad (F5)$$

which for normal atmospheric air at the temperature 20 °C can be given by

$$v_{Rey} = 1.50 \cdot 10^{-5} \frac{m}{s} \cdot \frac{1000mm}{\lambda}. \quad (F6)$$

The function $C_d(R_{Rey})$ is (almost) constant within the intervals given in the last column.